

POC ARCHITECTURE REVIEW

The DevOps OverSight Agent

AI-assisted incident response that correlates **Splunk** and **Datadog** over MCP to diagnose and remediate incidents — two reference architectures.

Solution 1 · Ballerina + MCP Proxy

Solution 2 · LangChain + A2A



Agenda

01

The problem

Why cross-platform incident diagnosis is slow, risky, and manual today.

02

Level 0 architecture

The concepts both solutions share: the mesh, telemetry fan-out, MCP, correlation, the human-approval gate.

03

Solution 1 — Ballerina + MCP Proxy

One agent, one MCP endpoint that federates Splunk & Datadog behind lazy-loaded tools.

04

Solution 2 — LangChain + A2A

An orchestrator that delegates to Datadog & Splunk specialist agents over the A2A protocol.

05

Comparison & path to enterprise

An honest scorecard, the 5-minute demo, and what production hardening looks like.

Executive summary

- **The pain:** in a P1, engineers burn the first 20–40 minutes hand-correlating Splunk logs against Datadog metrics — under pressure, by hand.
- **What we built:** an AI agent that runs that cross-platform investigation automatically, then proposes a specific fix and waits for a human to approve it.
- **Two architectures:** same POC, same 5-minute demo, built two ways to show the trade-off — a Ballerina MCP Proxy and a LangChain A2A multi-agent system.
- **The durable bets:** MCP as the contract boundary, correlation kept in one reasoning context, and propose-before-act as a hard rule — never autonomous.

≤ 5 min

inject → diagnose → approve → recover

< 90s

AI diagnosis (cloud model)

0

vendor creds to demo

◆ Bottom line

Both approaches work. This deck gives you the scorecard to choose.

01



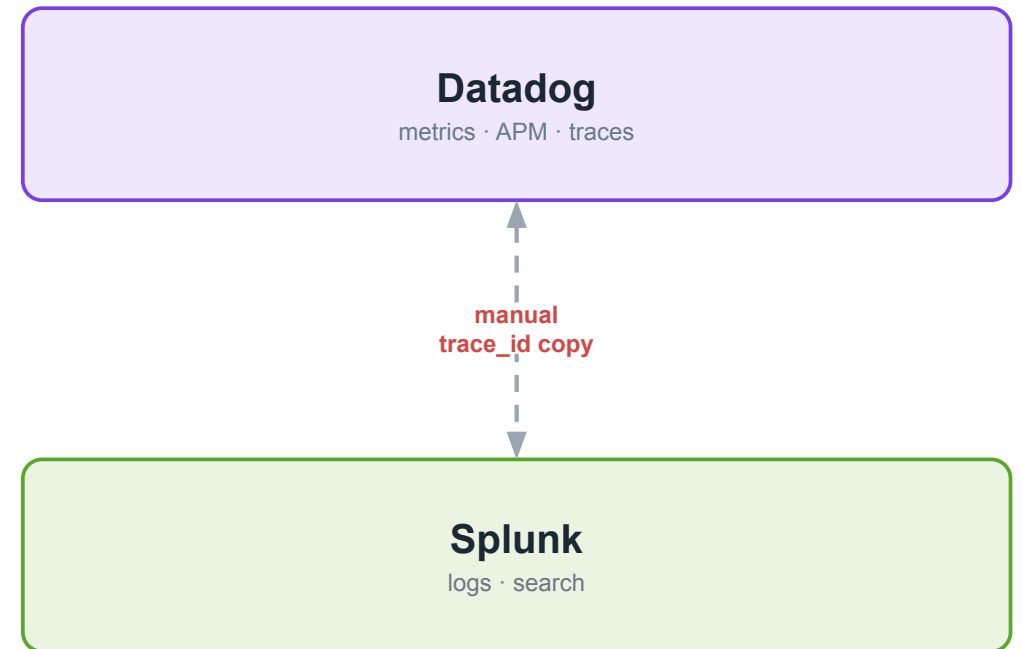
SECTION

The Problem

Why cross-platform incident diagnosis is slow, risky, and manual — and what 'good' looks like.

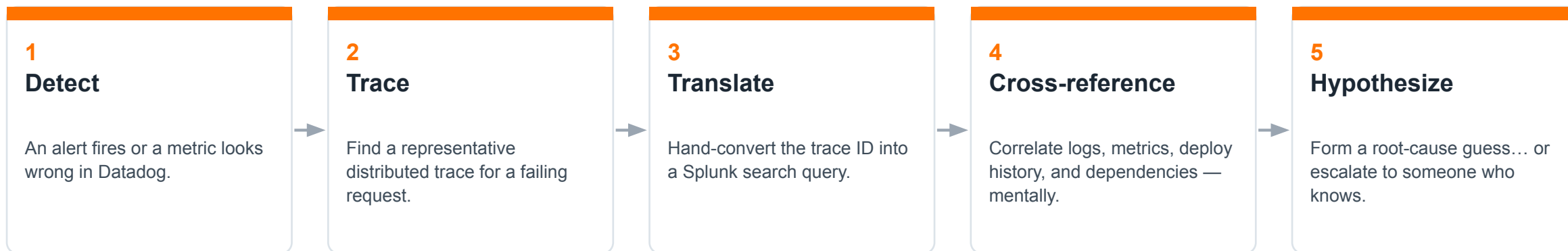
Best-of-breed tools, siloed by design

- **Two systems of record.** Logs and search live in Splunk. Metrics, APM, and traces live in Datadog. Both are excellent — and they don't share a query interface.
- **The join is manual.** During an incident an engineer swivel-chairs between them: spot a metric anomaly, pull a trace, copy the trace ID into a log search, cross-reference in their head.
- **It doesn't scale.** The work depends on individual institutional knowledge and gets harder as the mesh grows. Under pressure, at 2am, it's slow and error-prone.
- **Financial services raises the bar.** Any automation must be auditable, bounded, and must never change production without explicit human approval.



The engineer is the integration.

The manual-correlation tax



20–40 min
lost per major incident

◆ **Why this is the target**

Every one of these five steps is exactly what an LLM agent with the right tools can do — fast, consistently, and with its work shown.

Three forces shaped the design

Silo tax

Cut mean-time-to-diagnosis

Federate the two platforms so one investigation spans both — no context-switching, no manual trace-ID translation.

Approval gap

Middle path: assisted, not autonomous

Runbooks are auto-executed only after an explicit human approval. Enforced by the system, not by process or good intentions.

Lock-in risk

No AI vendor dependency

Model-agnostic by design — swap between a local model and cloud providers by config. Runs fully offline if needed.

Who we're building for

On-call / SRE

Primary user in a live incident

Plain-language root cause + blast radius + a specific action to approve — with evidence links back to both platforms.

Platform / Observability Eng

Operates & extends the system

Swap model or backend by config alone; the agent's own behavior is visible in the same stack it monitors.

Security & Compliance

Governance, change mgmt, audit

No state change without prior approval; a fixed action allowlist; every action recorded; secrets never in code.

Demo / Sales Engineering

Presents the capability

Full incident in ~5 minutes, offline, no vendor accounts, resettable to a clean state.

Service / App owners

Own individual services

Ownership, runbooks, and SLA metadata surfaced per service so response routes correctly.

What 'good' looks like — the POC bar

< 90s

AI diagnosis (cloud)

Under 3 min on a local model

≤ 5 min

Full incident cycle

Inject → diagnose → approve → recover

2

platforms, every time

Evidence from BOTH Splunk & Datadog

0

vendor credentials

Full cycle runs offline

◆ Propose-before-act

The agent must list available runbooks and receive an approval signal before it ever executes one. No exceptions.

◆ Model portability

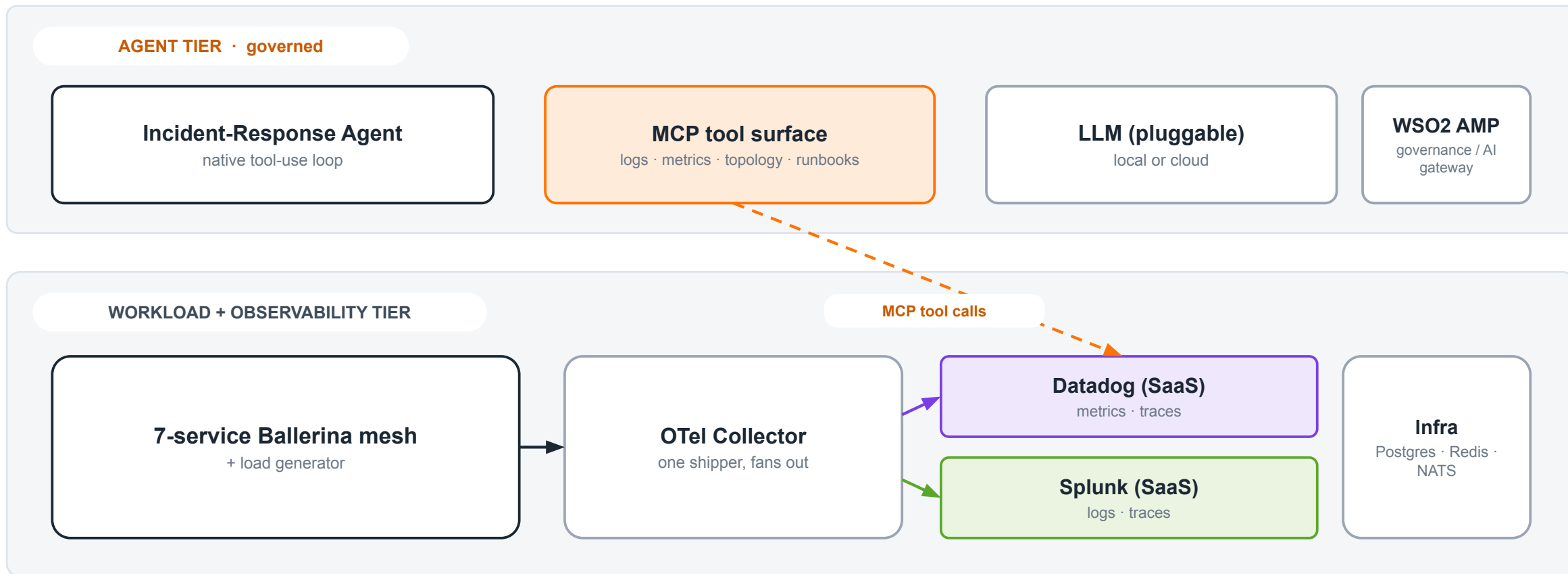
The same investigation must run on at least two providers (e.g. local Ollama and Anthropic) with no change beyond the provider selector.

SECTION

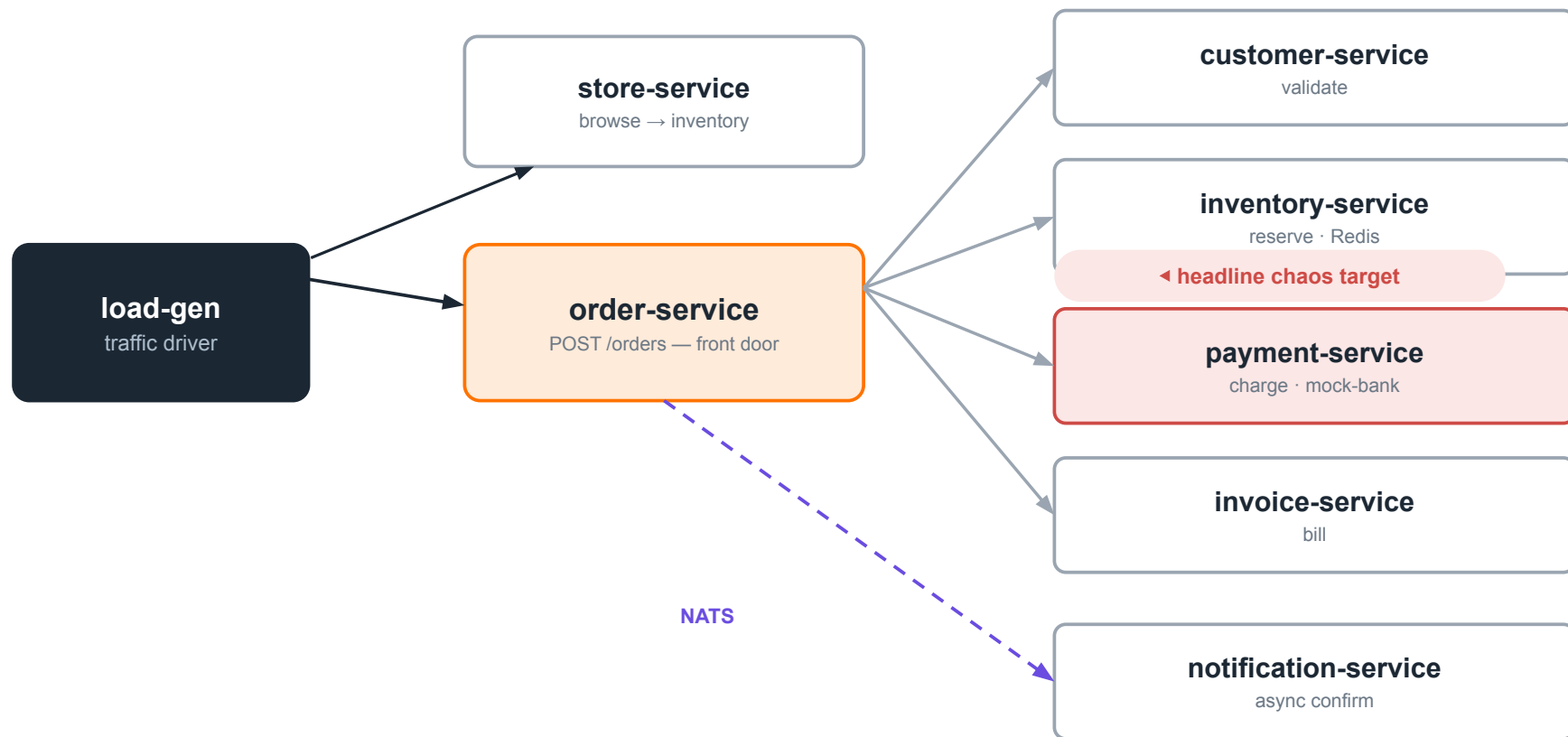
Level 1 Shared Architecture

The concepts both solutions share: the mesh, telemetry fan-out, MCP, correlation, and the approval gate.

The big picture: two tiers



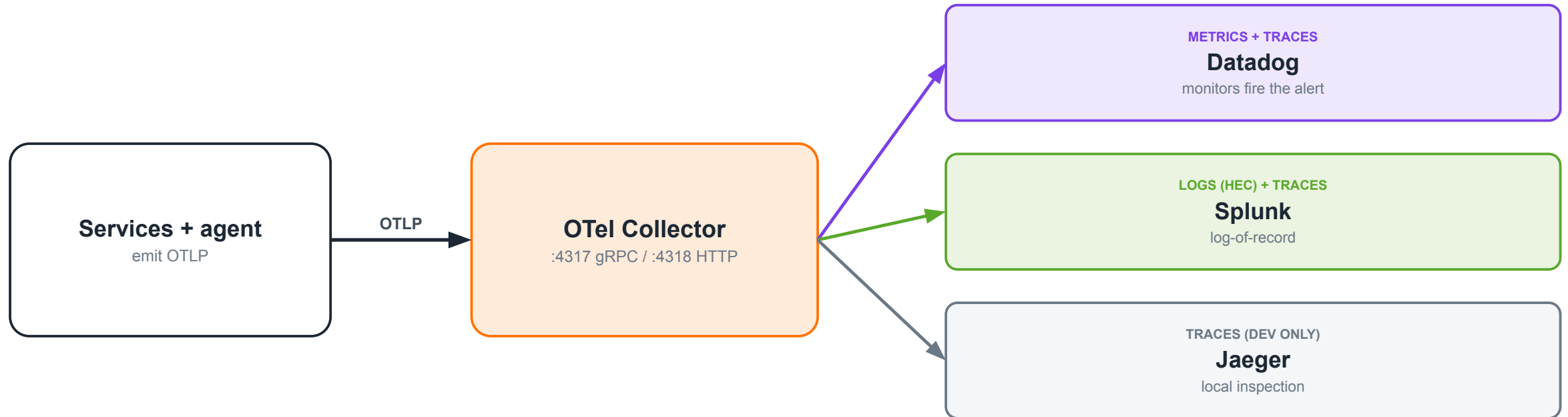
The mesh we observe & remediate



Why a mesh?

- Blast radius only exists across services.
- order fans out sync to 4 services + one async NATS hop to notification.
- Each service has a token-gated /chaos endpoint to inject faults.
- payment-service is the demo's failure point.

Telemetry fan-out: one collector, routed by signal

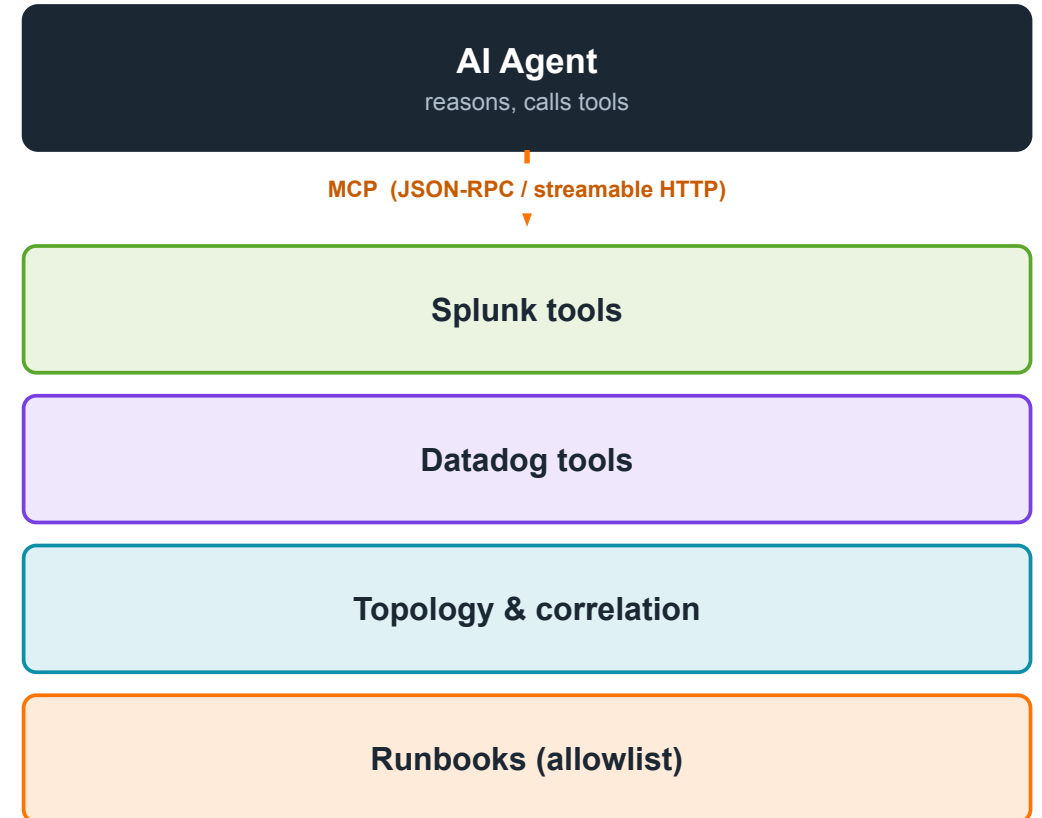


◆ The routing rule

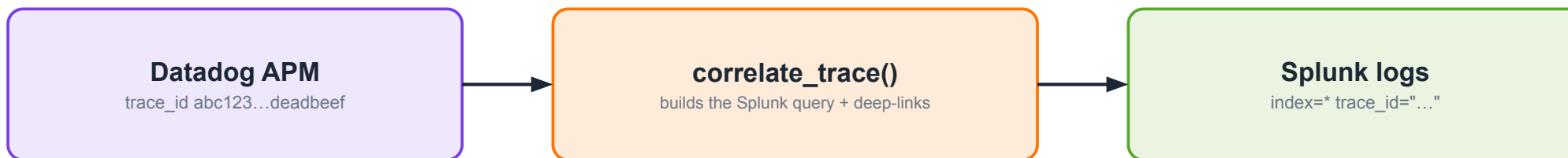
Traces → both platforms (so the agent can pivot by trace_id). Logs → Splunk only. Metrics → Datadog only. DD_LOGS_ENABLED=false avoids double-billing.

MCP: the contract boundary

- **Everything is a tool.** Splunk search, Datadog metrics, service topology, correlation, runbooks — the agent sees one uniform tool surface over MCP (Model Context Protocol).
- **Mock $\rightleftharpoons$ live is config, not code.** Swapping a mock backend for a real vendor MCP is a URL/token change. The agent's code never changes.
- **MCP fills what vendors can't.** Splunk MCP knows logs. Datadog MCP knows metrics. Neither knows YOUR catalog, dependency graph, owners, or runbooks — we add those as local tools.
- **One place for policy.** Routing, auth, and result hygiene get a single home at the boundary.



The core trick: correlate by trace_id



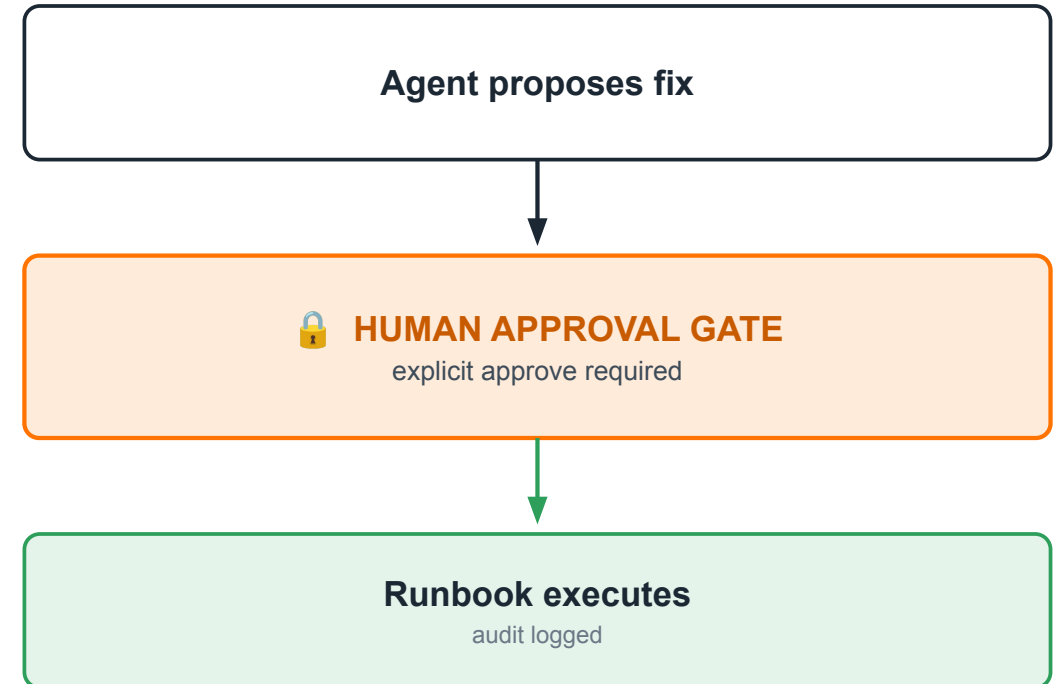
Both platforms carry the same trace_id in structured logs — that shared key is the whole game.

◆ ⚠ The #1 correctness gotcha — trace-ID width

Datadog surfaces a 64-bit dd.trace_id; OpenTelemetry & Splunk hold a 128-bit trace_id. Search Splunk with the wrong width and the agent wrongly concludes "no logs found." The correlation layer MUST normalize both. (The mock's single demo ID masks this — real traffic exposes it.)

The agent's job — and the gate

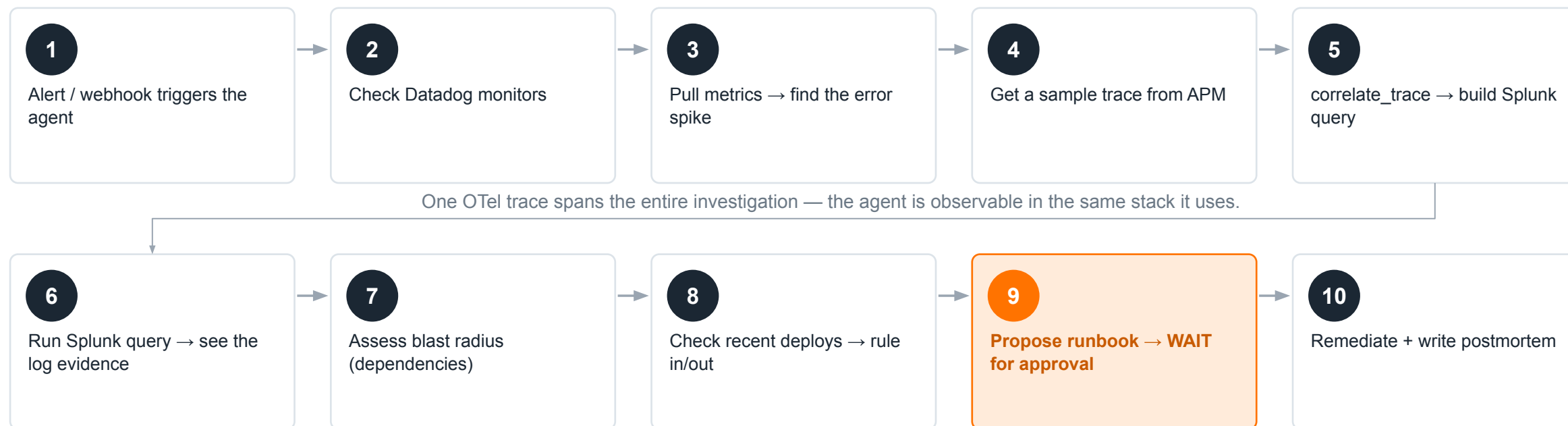
- **Investigate.** On an incident signal, run a structured cross-platform investigation in a native tool-use loop.
- **Synthesize.** Produce a root-cause summary with blast radius and evidence links into both Splunk and Datadog.
- **Propose.** Name a specific, vetted runbook and the parameters it would use — e.g. disable-chaos on payment-service.
- **Wait.** Stop. Do nothing to production until a human approves. This is the hard line.
- **Remediate & record.** On approval, run the runbook, stream progress, write an audit entry, and post a postmortem.



◆ Bounded

Only a fixed, pre-approved runbook list. No arbitrary command execution — ever.

End-to-end: the 10-step investigation



WSO2 Agent Manager's role

◆ Runtime & governance

Hosts the agent as a platform-managed workload from a Git project. Lifecycle, identity, and policy live here.

◆ AI gateway

All LLM traffic can route through AMP for rate-limiting, quota, model routing, and model-level audit — injected by config.

◆ Self-observability

The agent's own spans — reasoning, tool calls, token use, latency — flow through the same collector. Detect runaway agents.

◆ Honest scoping for tonight

In Solution 1 (Ballerina) AMP is the agent runtime. In Solution 2 (LangChain) the agents run as plain processes and AMP appears only as an optional LLM gateway. AMP is the enterprise governance plane — not a hard dependency of the POC.

No lock-in: the model is a config flag

Ollama

LOCAL · CREDITS-FREE

qwen2.5 / qwen3.5 family

On-prem, zero cost, offline demo

Good for development/unit testing.

Testing proved that this did not function at a production level response level.

Anthropic

CLOUD

claude-sonnet-4-6

Fastest, highest-quality tool use

OpenAI

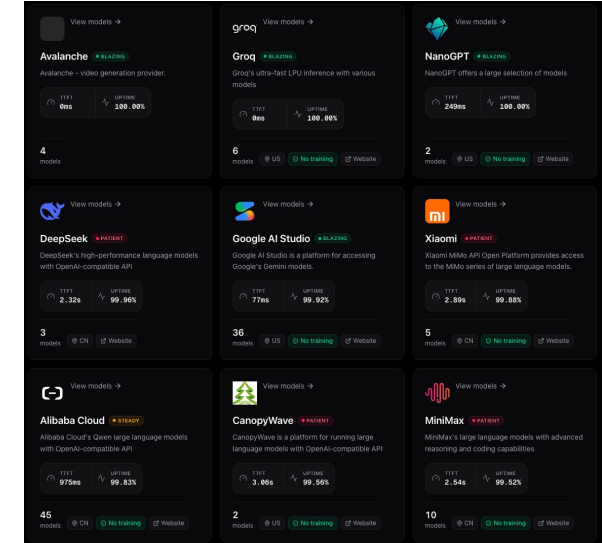
CLOUD

gpt-4o

Alternative managed provider

Other LLM Vendors...

CLOUD



Switch with one variable: **LLM_PROVIDER = ollama | anthropic | openai**

The default demo runs fully offline on a local model — no vendor account required; however it does not perform at a production level

THE FORK

Two roads to the same goal

Everything so far is shared. Here's where the two solutions diverge — and it comes down to one question:

Where do the topology/correlation tools live, and how do we keep the agent's context small?

SOLUTION 1

Ballerina + MCP Proxy

One agent talks to one MCP Proxy that federates the backends and lazy-loads their tools. Low-context by a server-side registry.

SOLUTION 2

LangChain + A2A

An orchestrator delegates to specialist agents over A2A. Low-context by decomposition — each specialist owns only its own small toolset.

SECTION

Ballerina + MCP Proxy

One agent, one MCP endpoint that federates Splunk & Datadog behind lazy-loaded tools.

The core idea

- **One agent, one endpoint.** The Ballerina agent connects to exactly one MCP server — the MCP Proxy on :8290. It never touches Splunk or Datadog directly.
- **The proxy federates.** Behind that one endpoint it fans out to the Splunk and Datadog MCP backends and adds local topology, correlation, and runbook tools.
- **Small context by lazy loading.** The proxy hides the big vendor tool manifests in a server-side registry and reveals only what the agent asks for.
- **Full-stack Ballerina.** Agent, proxy, mock backends, and the 7-service mesh are all Ballerina 2201.13.3 — one language, one build.

◆ The best-practice claim

Cross-system correlation is a single-context reasoning task. Keep it in ONE agent — federate the tools, don't fragment the reasoning.

1

MCP endpoint the agent sees

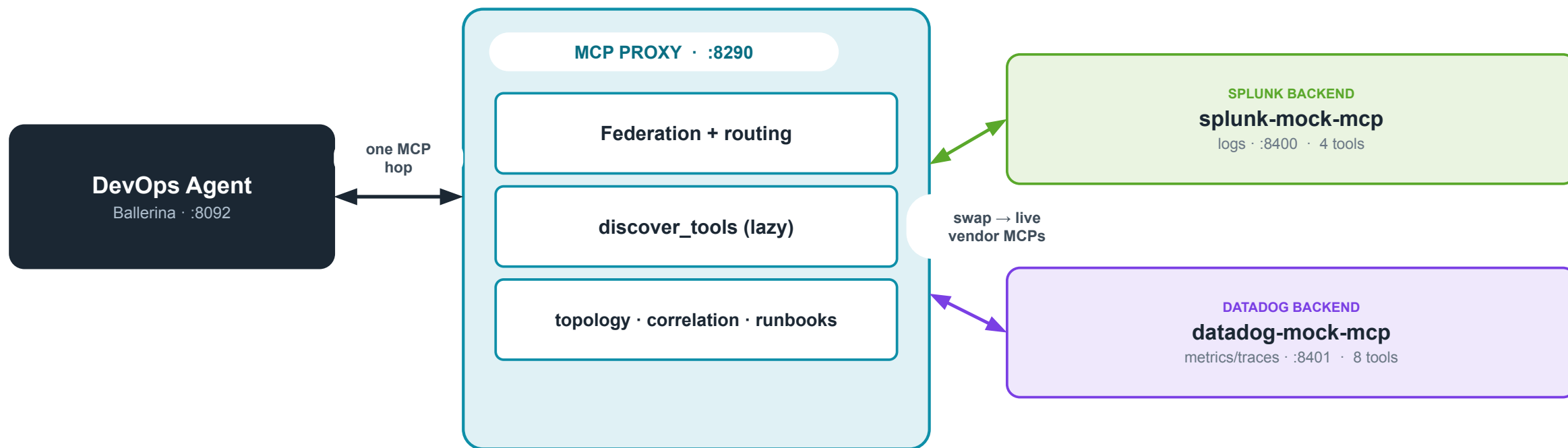
12

topology tools

1

language

POC Architecture



Why a proxy?

Small agent context

The vendor MCPs expose 50+ tools each. Injecting all of them would blow the context window. The proxy keeps them server-side.

Mock $\neq$ live isolation

One env var on the proxy switches mock to SaaS. The agent is untouched, so you can stage a rollout safely.

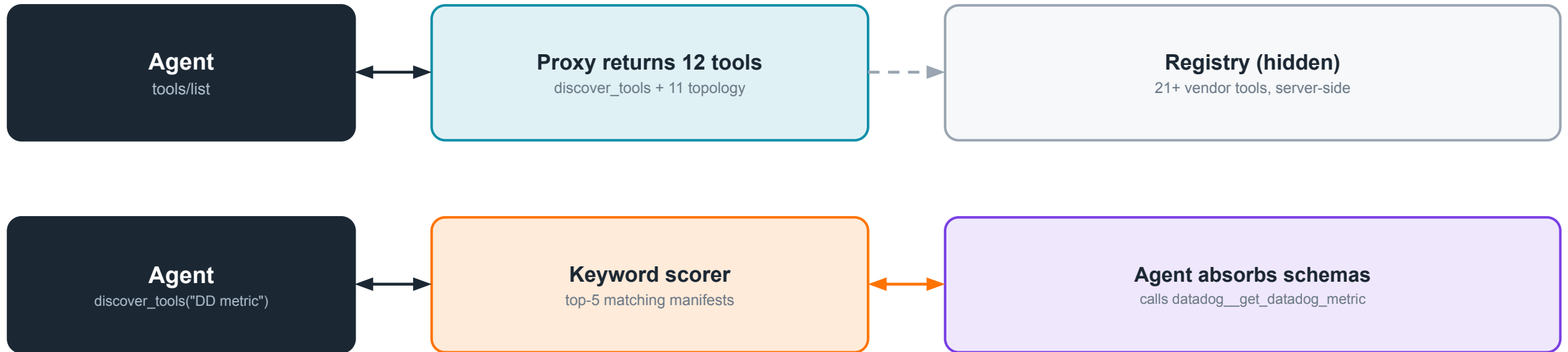
Single policy chokepoint

Routing, auth (via a gateway), and result hygiene get exactly one home instead of being smeared across the agent.

Fills the vendor gap

It's Ballerina, so it can also ACT — hit chaos endpoints, run runbooks — and it owns the catalog the vendors don't have.

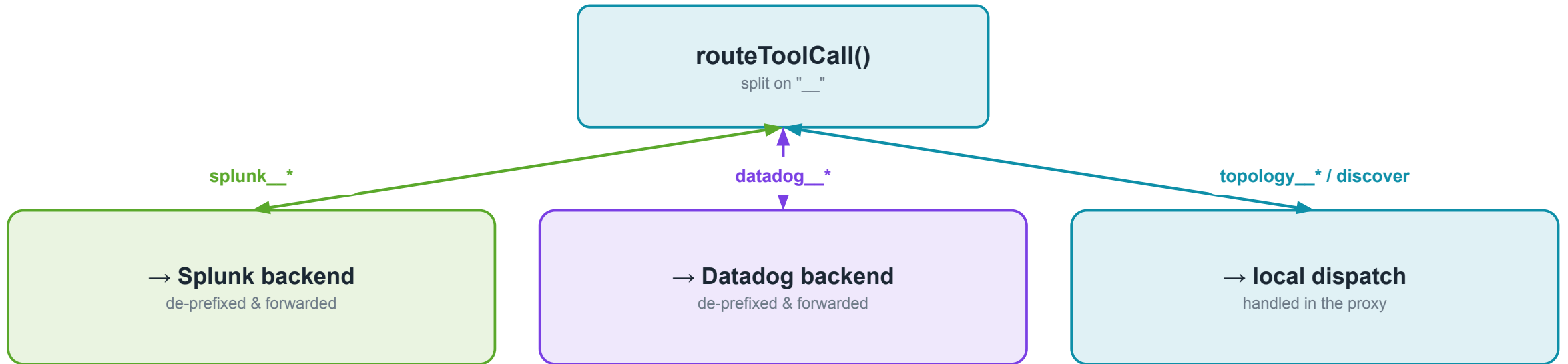
Lazy tool loading — the low-context pattern



◆ POC honesty

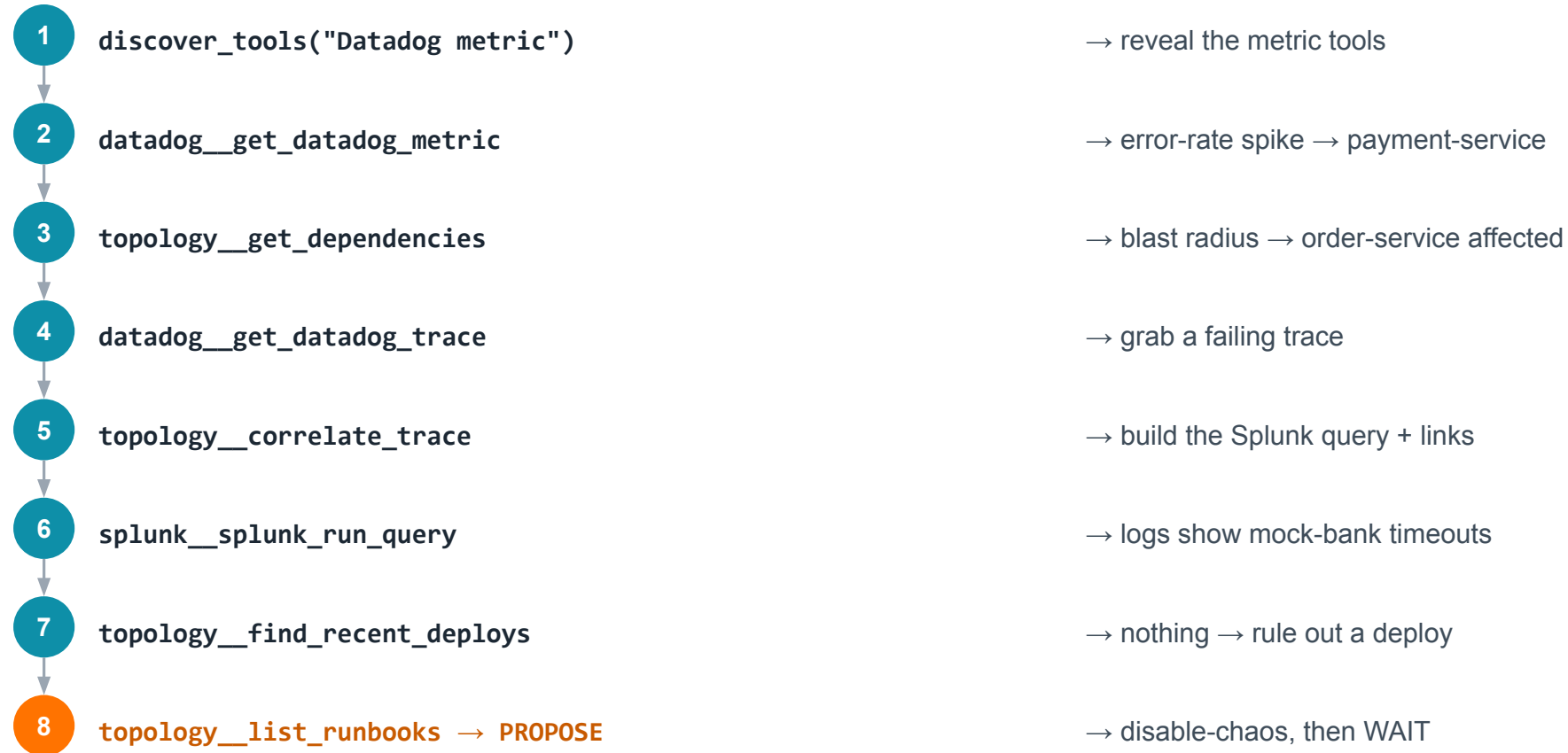
Today the router is a keyword scorer — accurate enough at ~21 tools. Production swaps in a pgvector + embedding semantic router when the live MCPs bring 50+ tools each.

Prefix routing keeps it simple



What the agent can reach: **11 topology__** (lookup / dependencies / health / correlate_trace / deploys / incidents / runbooks / audit) + **4 splunk__** + **8 datadog__**

The investigation loop in action



Stack, tools & runbooks

Layer	Choice
Language / runtime	Ballerina Swan Lake 2201.13.3
MCP transport	Streamable HTTP · JSON-RPC 2.0
Telemetry	ballerinax/jaeger (OTLP) + prometheus
LLM loop	native tool-use, no SDK · maxTurns 30
Default model	Ollama qwen (local) / claude-sonnet-4-6

Runbooks — the fixed allowlist

- disable-chaos** LIVE — POST /chaos/reset. The demo's recovery lever.
- restart-service** stub — kubectl rollout restart
- clear-cache** stub — Redis FLUSHDB on inventory
- freeze-deploys** stub — sets an in-memory flag

Every execution appends to an in-memory audit log. No run_cli, ever.

Strengths & honest limits

✓ Strengths

- **Small, stable context** no matter how many tools the real MCPs expose.
- **Correlation in one context** logs + spans + topology join in one working set.
- **Mock $\nrightarrow$ live is config** one env var, proxy-side.
- **Single policy chokepoint** routing, auth, hygiene in one place.
- **Bounded remediation** typed runbook allowlist + approval gate.
- **Clean migration path** wrap with A2A at org boundaries later.

⚠ Known limits (POC)

- **Keyword router** not vector-based; fine at ~21 tools, upgrade for 50+.
- **No result hygiene** live payloads need truncation/neutralization.
- **No endpoint auth** trusted local net; prod defers to the WSO2 gateway.
- **Static catalog** in-code map today; production reads a CMDB.
- **Stub runbooks** only disable-chaos is wired to a real action.
- **SSE buffering risk** a buffering gateway breaks streaming run_runbook.

SECTION

LangChain + A2A

An orchestrator that delegates to Datadog & Splunk specialist agents over the A2A protocol.

The core idea

- **Three agents.** An orchestrator (DevOpsOverSightAgent) delegates to two specialists — a DataDogAgent and a SplunkAgent — over the A2A protocol.
- **Low-context by decomposition.** Each specialist eagerly loads only its own small toolset (8 Datadog / 4 Splunk). The vendor manifests never enter the orchestrator's context.
- **No proxy, no discover_tools.** The proxy dissolves: federation becomes the A2A boundary; correlation & runbooks stay in-process in the orchestrator.
- **Familiar ecosystem.** Python 3.12, LangChain 1.x on the LangGraph runtime — the stack most teams already know.

◆ Same principle, other shape

A2A is used ONLY at the platform-team boundary — not to split correlation. The fusion still happens in one orchestrator context.

3

agents

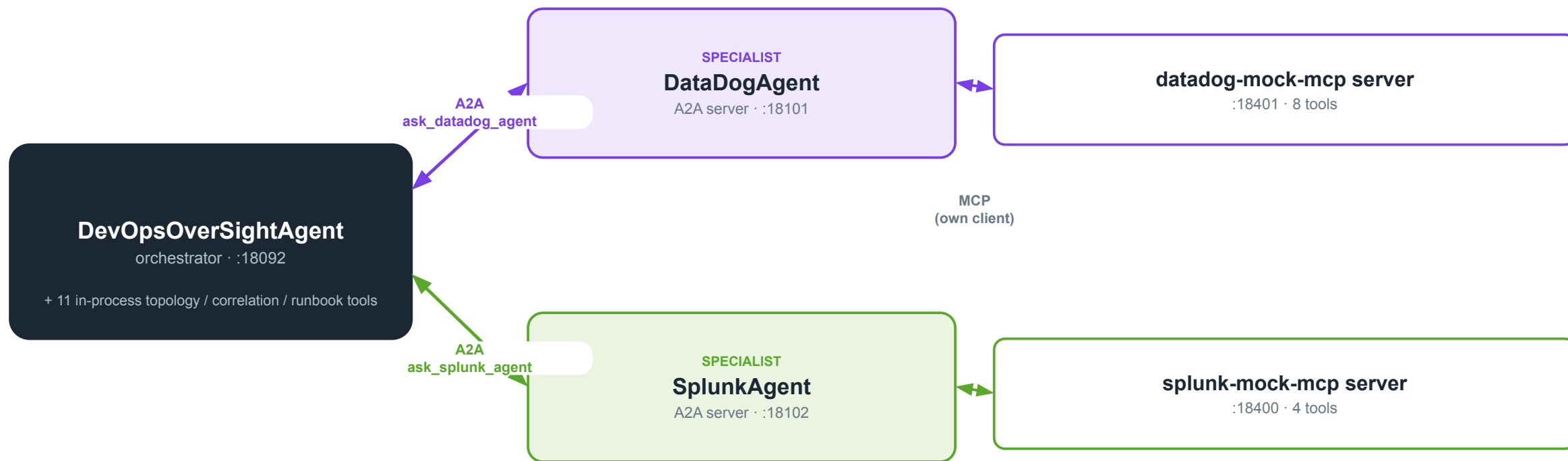
8+4

specialist tools

Python 3.12

LangChain 1.x · LangGraph

POC Architecture



The A2A protocol

- **Agent cards.** Each specialist publishes `/.well-known/agent-card.json` declaring its skill (`datadog_evidence`, `splunk_log_search`) and capabilities.
- **JSON-RPC delegation.** The orchestrator resolves the card, then sends a message and streams the reply. One-way: orchestrator → specialist.
- **Request/response, not long-running tasks.** Each call is a single round-trip — verified against the SDK, no Task lifecycle.
- **The output contract is load-bearing.** Evidence crosses as prose, so specialists MUST return `trace_ids` verbatim, metric values, timestamps. A vague reply starves the orchestrator.

The delegation call

orchestrator LLM

```
→ ask_datadog_agent("...")  
→ A2ACardResolver.get_card()  
→ SendMessageRequest (JSON-RPC)  
→ specialist ReAct loop → MCP
```

◆ Version pin matters

`a2a-sdk ≥1.1, <2` is protobuf-based. The older 0.2.x tutorials do NOT apply — build against the installed SDK.

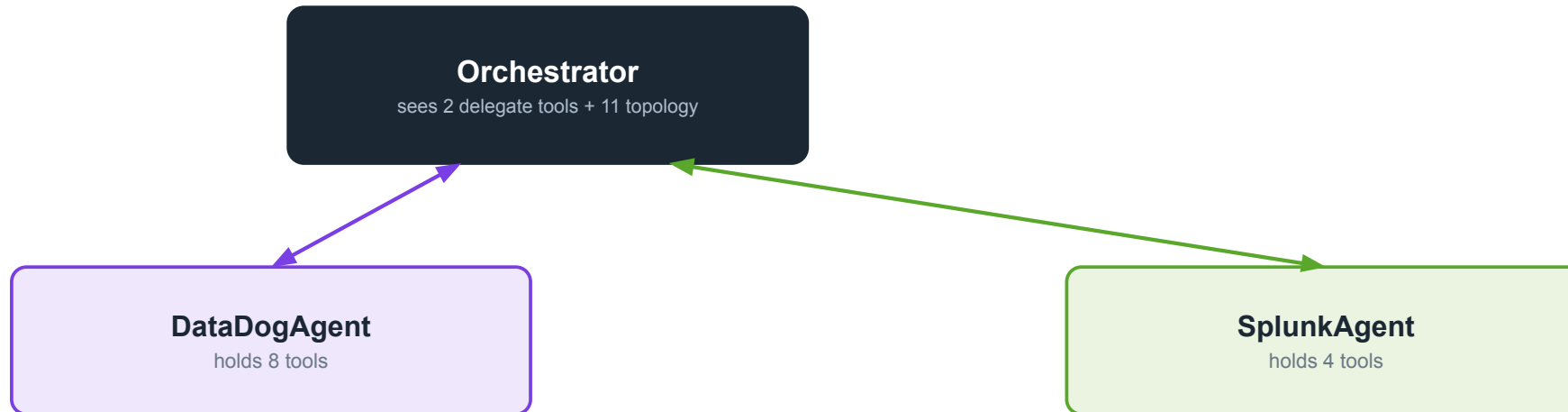
Low-context by decomposition

The problem (shared)

Real vendor MCPs expose 50+ tools each. Load them all into one agent and the context window is gone.

S2's answer

Give each platform its own agent. The orchestrator only ever sees 2 delegate tools, never the vendor manifests.

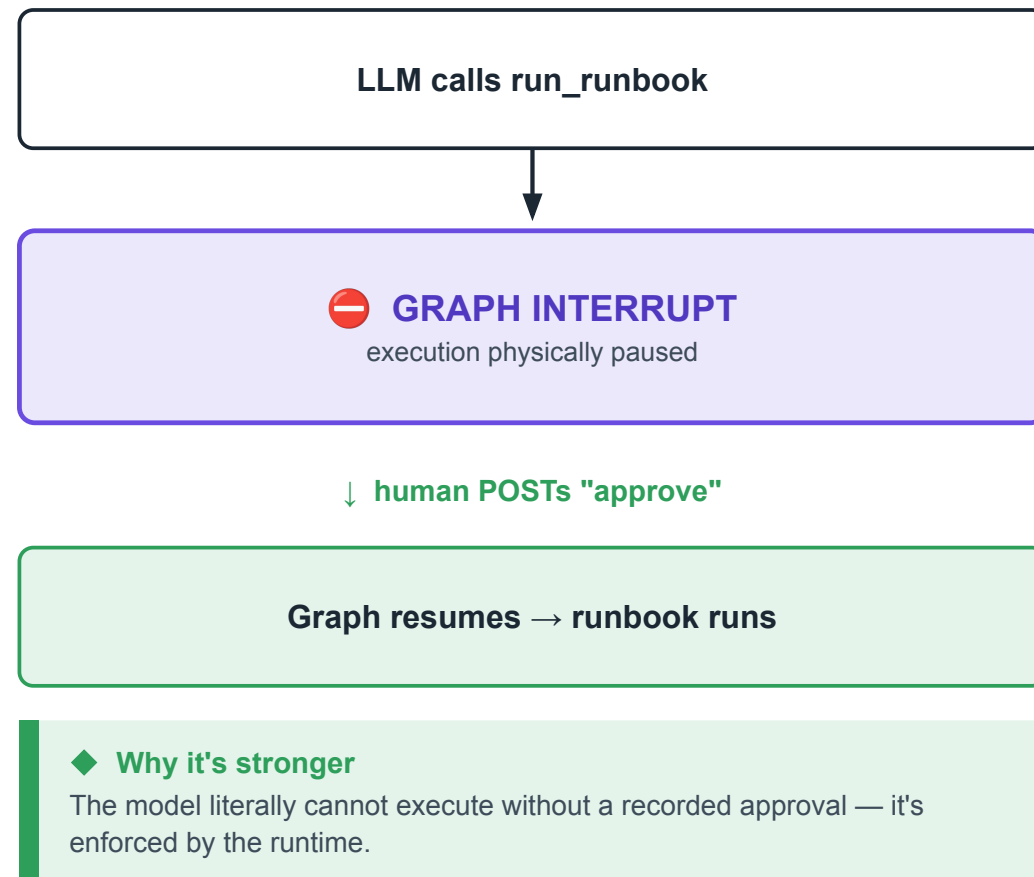


◆ Trade

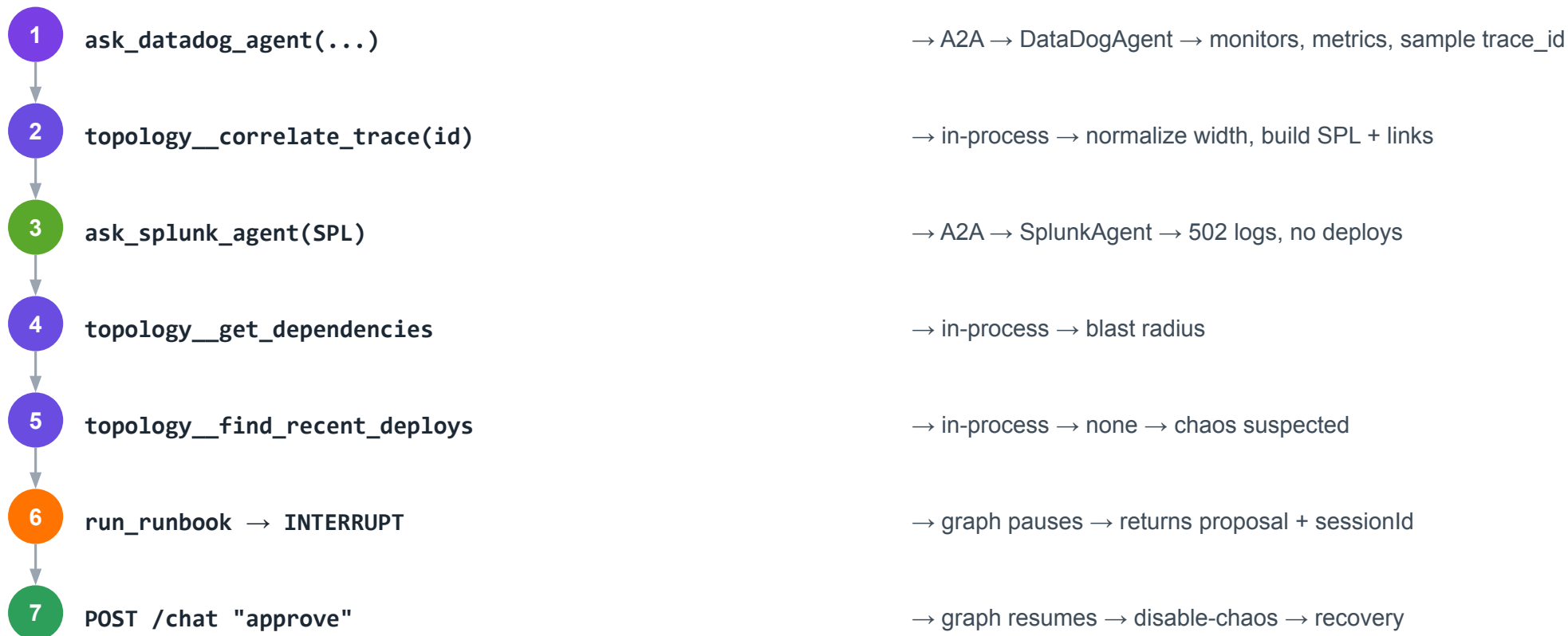
Cleaner separation & familiar code — at the cost of another network hop and 3× the model non-determinism.

The hard approval gate

- **Code-level, not prompt-level.** A LangGraph `HumanInTheLoopMiddleware` physically interrupts the graph before `topology__run_runbook` executes.
- **The graph pauses.** `/investigate` returns the proposal + a `sessionId`. State is checkpointed by `InMemorySaver` keyed to that session.
- **Human resumes it.** Operator POSTs "approve" to `/chat`; the graph resumes with a `Command(resume=...)` and only then runs the runbook.
- **Fail-safe by default.** Any non-approval reply is treated as rejection. The runbook does not run.



The investigation loop with A2A

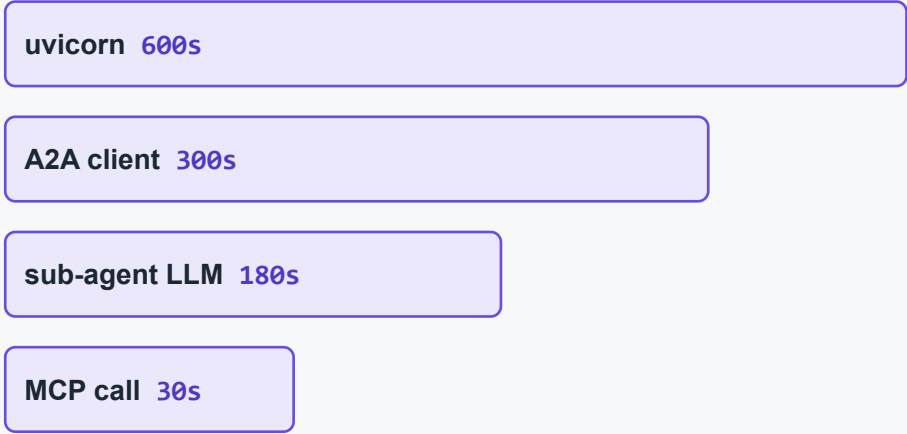


A2A delegation · in-process tool · gate

Stack & the Timeout Chain

Layer	Choice
Language / runtime	Python 3.12 · single uv workspace
Agent framework	LangChain 1.x on LangGraph 1.x (create_agent)
A2A	a2a-sdk ≥1.1,<2 (protobuf)
MCP	official mcp SDK · langchain-mcp-adapters
Gate / state	HumanInTheLoopMiddleware + InMemorySaver

The Timeout Chain (must stay ordered)



Largest-outermost. assert_timeout_chain() fails fast at startup if misordered.

Strengths & honest limits

✓ Strengths

- **Familiar ecosystem** Python + LangChain; low ramp for most teams.
- **Hard, code-level gate** graph interrupt, not a prompt instruction.
- **Structural low-context** specialists cap tools by construction.
- **Clean protocol separation** A2A at the boundary, MCP for tools, in-process fusion.
- **Same model flexibility** 4 providers, config-swappable.
- **Full self-observability** one trace spans user → A2A → specialist → MCP.

⚠ Known limits (POC)

- **3× model non-determinism** three agents must each emit clean tool calls.
- **Prose output contract** vague specialist replies starve correlation.
- **Extra network hop** A2A adds latency vs an in-process call.
- **Timeout Chain to maintain** four nested timeouts, order-sensitive.
- **In-memory state** restart mid-approval drops the pending runbook.
- **SDK version drift** a2a-sdk 1.1 is protobuf; pin carefully.

SECTION

Comparison & Path Forward

An honest scorecard, the 5-minute demo, and what production hardening looks like.

The architectural crux

Both keep correlation in ONE reasoning context. They differ only in how they keep the toolset small.

Solution 1 — federate the tools

- **Low-context via** a proxy with a server-side registry + lazy discover_tools.
- **Correlation locus** one Ballerina agent holds all evidence directly.
- **Boundary** one process federates; nothing is fragmented.

Solution 2 — decompose the agents

- **Low-context via** specialist agents that each own a small toolset.
- **Correlation locus** the orchestrator fuses prose evidence from specialists.
- **Boundary** A2A at the platform-team line — NOT inside correlation.

What we actually measured

Metric — measured on a production-grade AI model	Ballerina + Proxy	LangChain + A2A
Correct diagnosis rate	100% (18 / 18 test runs)	100% (18 / 18 test runs)
Median time to a proposed fix	~14 seconds	~36 seconds
Test runs evaluated	18 (3 independent batches of 6)	18 (3 independent batches of 6)

◆ Why this matters

Both are reliable enough to trust in a live environment. Ballerina resolves faster in this test because it makes fewer hops between agents for the same investigation.

◆ A critical cost finding

A free, locally-hosted AI model is possible for either approach, but was only 28–61% reliable in the same test. Reliability came from using a production-grade model, not from which architecture we chose — worth weighing directly against licensing cost.

Method: identical simulated incident injected into a test service; timed end-to-end from alert to a proposed fix; 3 independent test batches of 6 runs per approach. Testing cost using Claude Haiku 4.5: \$2.56.

Side-by-side scorecard

Dimension	Solution 1 · Ballerina + Proxy	Solution 2 · LangChain + A2A
Low-context strategy	Proxy lazy-load (server registry)	Agent decomposition (specialists)
Correlation locus	One agent, direct	Orchestrator fuses prose
Approval gate	Prompt-level instruction	Code-level graph interrupt
Language / stack	Ballerina 2201.13.3	Python 3.12 / LangChain 1.x
Moving parts	Fewer — 1 agent + proxy	More — 3 agents + hop
Model non-determinism	×1 (single agent)	×3 (each specialist)
Ecosystem familiarity	Ballerina — niche	Python/LangChain — broad
Tool-sprawl at 50+ tools	Needs vector router upgrade	Handled by decomposition
WSO2 AMP fit	Native agent runtime	Native agent runtime
Ops surface	Single reasoning process	Distributed; timeout chain

Recommendation

◆ **PLACEHOLDER — to complete together after the scorecard discussion**

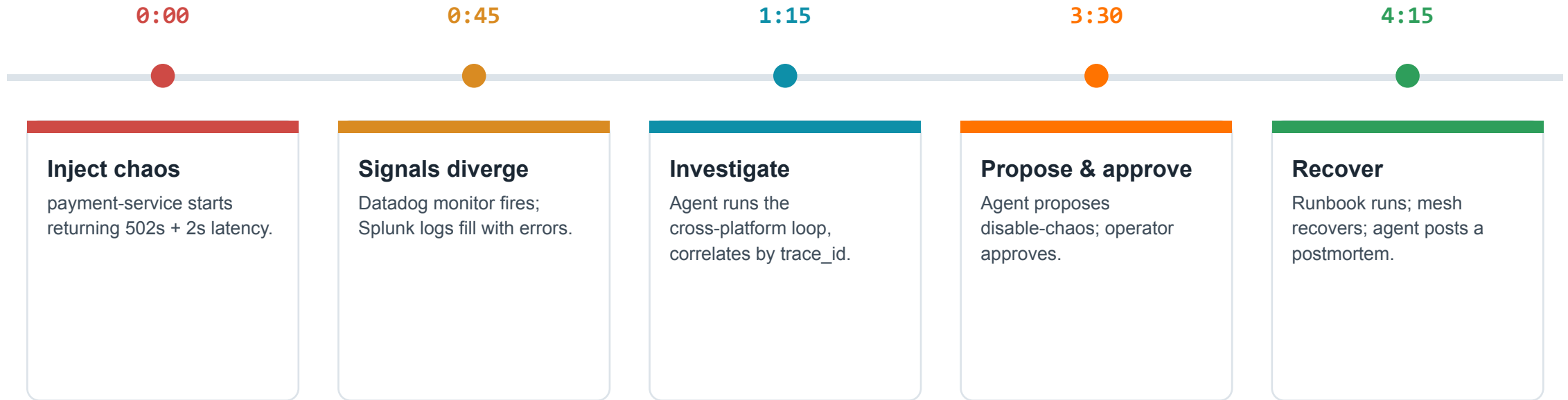
■ **Lean Ballerina + Proxy if...**

- You're standardizing on WSO2 AMP as the agent platform.
- You want the fewest moving parts to operate and audit.
- Single-context correlation fidelity is paramount.
- A niche language is acceptable for a small platform team.

■ **Lean LangChain + A2A if...**

- Your teams already live in Python / LangChain.
- A code-enforced approval gate is a compliance must-have.
- You expect many platforms / trust domains (A2A scales out).
- You're comfortable operating a distributed agent system.

The 5-minute demo — payment-service P1



◆ Runs offline, resets clean

No vendor credentials. A token-gated chaos endpoint injects the fault; disable-chaos (or reset) returns to a clean baseline for the next run.

POC → Enterprise: the hardening path

Capability	POC state	Enterprise requirement
Identity & access	Unauthenticated local net	OIDC/SSO + workload identity + per-tool RBAC at the gateway
Secrets	.env variables	Vault / cloud secrets manager; rotation
Audit log	In-memory, session-scoped	Durable, tamper-evident, tied to identity, SIEM-fed
Approval gate	S1 prompt / S2 code-level	Hard code-level interception in both; recorded signal
Service catalog	Static in-code map	CMDB-backed, team-maintained, versioned
Observability backends	Mock MCP servers	Live Splunk (app 7931) + Datadog (Bits AI) MCPs
Trace-ID handling	Verbatim (demo id)	True 64/128-bit normalization on real traffic
Scale	Single node	Agents on K8s; proxy behind LB; managed infra

Cross-cutting gotchas to respect

Trace-ID width (64 vs 128-bit)

The #1 correctness bug. Normalize or the agent sees 'no logs.' Mocks mask it.

NATS async propagation

HTTP trace context auto-propagates; NATS does not. Inject traceparent into the envelope.

Untraced Postgres

Turn on SQL connector tracing or DB latency is invisible to the agent.

SSE buffering at the gateway

A buffering proxy breaks streaming run_runbook progress.

Model non-determinism (Ollama)

Keep maxTurns ≥ 25 ; prefer a cloud model for the live demo.

In-memory state

Audit log & pending approvals are volatile — a restart drops them.

Summary & next steps

What we proved

- **The capability works.** cross-platform diagnosis + approved remediation in a 5-minute, offline, laptop demo.
- **Two credible architectures** — Ballerina + MCP Proxy and LangChain + A2A — each with clear trade-offs.
- **The durable bets hold:** MCP as the contract, single-context correlation, propose-before-act, model portability.
- **A clean path to production** exists for either — a hardening checklist, not a re-architecture.

NEXT STEPS

- 1 Fill in the recommendation with the client's team & governance context.
- 2 Pick one architecture for a hardening spike (auth + live MCPs).
- 3 Wire real Splunk & Datadog MCPs; exercise trace-ID normalization.
- 4 Run the demo live with the client on their own laptop.

Ports & endpoints reference

Solution 1 · Ballerina	Port
DevOps Agent	8092 → 8000
MCP Proxy	8290
splunk-mock-mcp	8400
datadog-mock-mcp	8401
payment chaos endpoint	9196
Prometheus scrape	9797

Solution 2 · LangChain	Port
Orchestrator	18092 → 8000
DataDogAgent (A2A)	18101
SplunkAgent (A2A)	18102
datadog-mock-mcp	18401
splunk-mock-mcp	18400
OTLP collector	14317 / 14318

Shared: demo trace_id **abc123def456789012345678deadbeef** · WSO2 AMP console :**3000** · chaos token **dev-chaos-token**

The LangChain stack 1-prefixes every host port so both stacks run side-by-side for a live A/B on one machine.

06

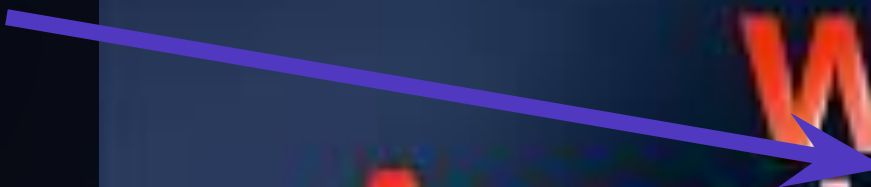


SECTION

Reference Product Stack

WSO2 Agent Manager

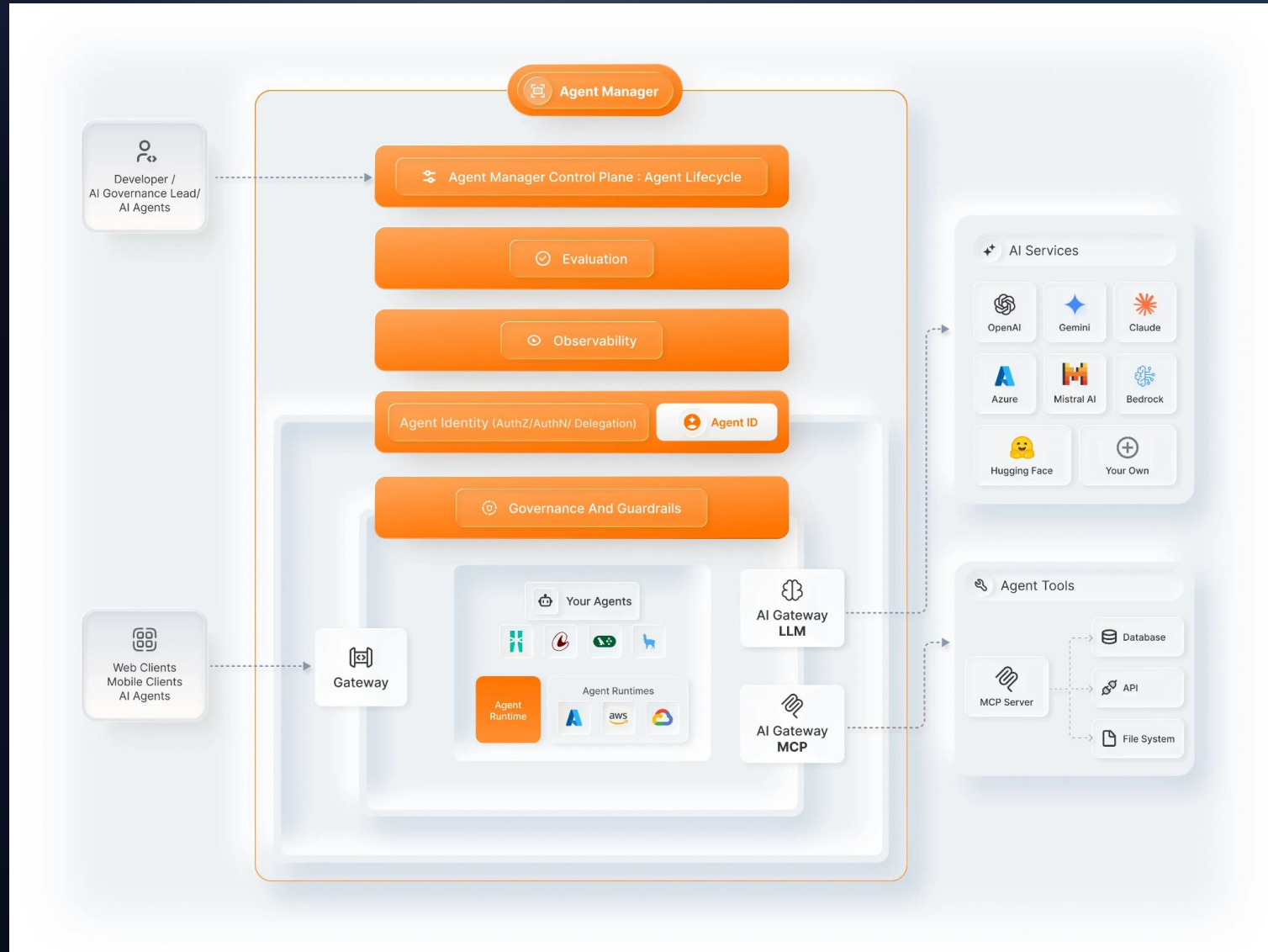
Click
Me



WSO2
Agent Manager



WSO2 Agent Manager Architecture

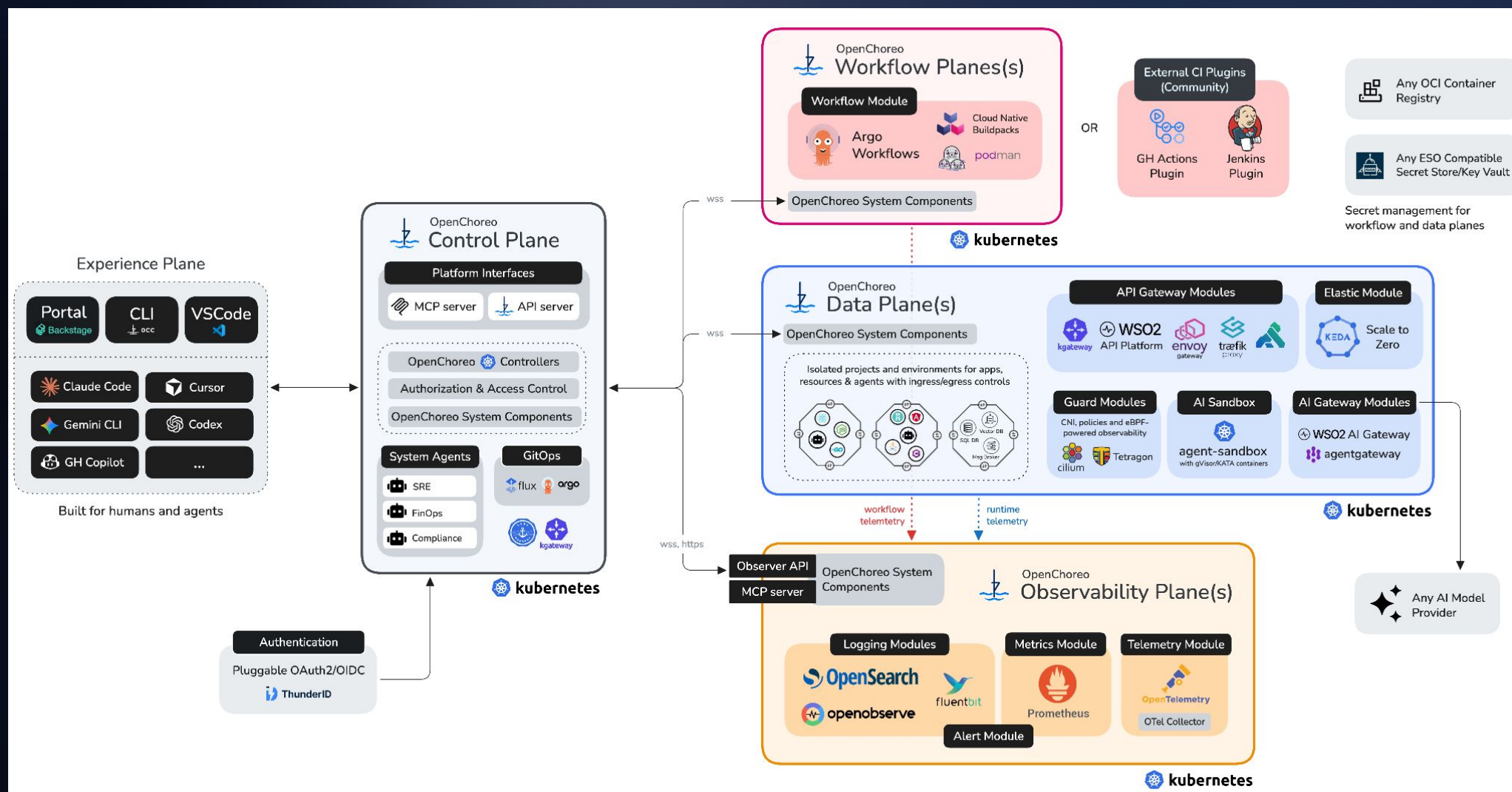


OpenChoreo

Click
Me



OpenChoreo Architecture



07

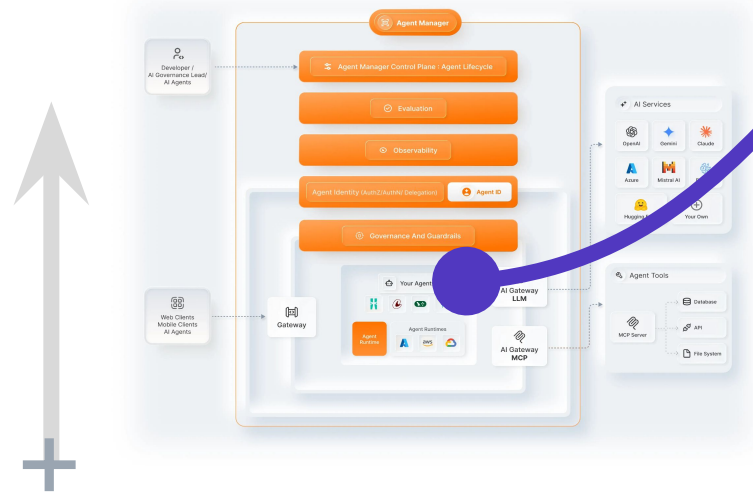


SECTION

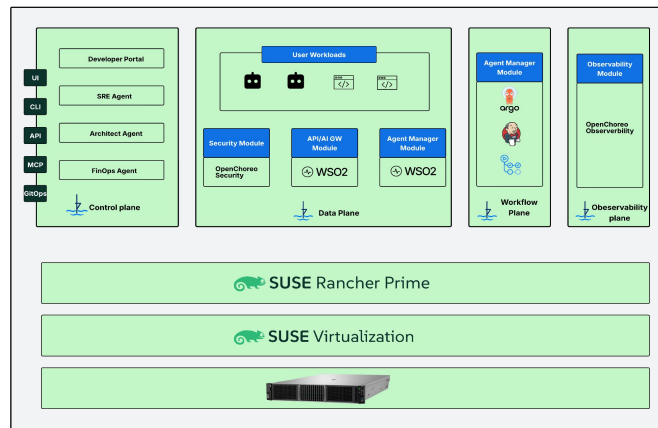
Reference Deployment Architecture

Putting It All Together

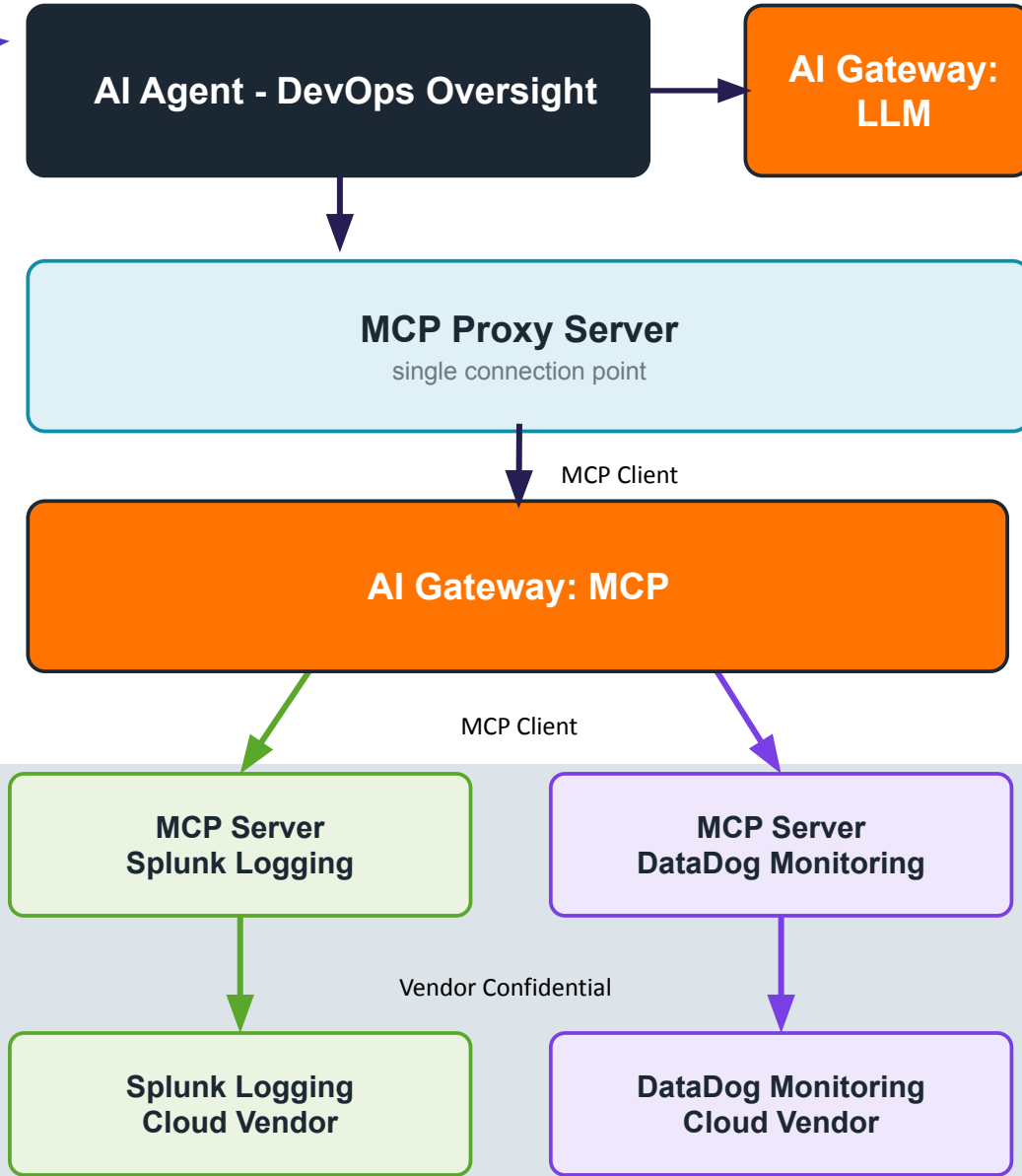
WSO2 Agent Platform: Agent Manager



OpenChoreo on SUSE Rancher



Agent Platform + DevOps Oversight Agent



Thank you.

Questions, pushback, and better ideas all welcome — this is a team effort.

Scott Bechtel · WSO2 Forward Deployed Engineering

